

pySnowClim: fast process-based snow modeling in Python

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Summary

pySnowClim is an open-source Python implementation of the process-based SnowClim model for simulating snow dynamics including accumulation and melt processes. The model achieves a balance between representing important physical processes, such as energy balance calculations, snow density evolution, albedo dynamics, and phase change processes, while simplifying other components to maintain computationally efficiency, delivering consistent snowpack simulations across diverse climatic conditions. Building on these strengths of SnowClim, pySnowClim adopts a layered architecture with vectorized multi-point computation using standardized NetCDF I/O. This makes pySnowClim suitable for point to large scale applications, and ready to couple with climate and hydrological models.

Statement of need

Snow is a critical component of the global water cycle. The accurate simulation of snowpack dynamics is not only essential for water resource management, but also for flood prediction, ecological studies, and climate change impact assessments ([J. T. Anderson, 2023](#); [J. T. Anderson et al., 2025](#); [Dixon et al., 2026](#); [Williams et al., 2026](#)).

The target audience of pySnowClim includes hydrologists, climatologists, ecologists, water resource managers, and students who need reliable snow modeling capabilities for applications such as:

- Research:** Detailed energy balance studies and process investigations
- Operations:** Water resource forecasting and management
- Education:** Teaching snow physics and energy balance concepts
- Climate Studies:** Long-term snow evolution under changing conditions
- Adaptation Planning:** Anticipating and planning adaptations to impacts of warming on snow-dependent species, ecosystems, and communities

In addition, the model can be used on different:

- Spatial Scales:** Point locations to continental domains
- Temporal Scales:** Sub-daily to daily timesteps, multi-decadal simulations
- Environments:** Diverse snow climates from maritime to continental

State of the field

Many current snow models are either computationally efficient but only represent physical processes to a very limited extent (e.g. temperature index models), or provide detailed physical process representation but are too computationally burdensome for large-scale high-resolution

applications (e.g. most process-based models). The former category includes temperature index models such as SNOW-17 (E. A. Anderson, 1973) which is used by the US National Weather Service for river flow and flood forecasting, and the snow components of the SRM (Martinec et al., 2008), HBV (Bergstrom, 1992), and SWAT (Arnold et al., 1998) models which are used for water management, flood forecasting, and climate change studies. These models often only require temperature and precipitation for the snow routines and can be applied over regional to continental domains and multidecadal time periods, but may not be robust under changing climatic conditions. This is because the temperature index method implicitly assumes that air temperature as a proxy for the energy available to melt snow and the use of static degree-day factor cannot represent the shift in energy fluxes associated with earlier snowmelt in a warmer climate (Ismail et al., 2023) and energy fluxes that cause rain-on-snow events that could lead to extreme flooding (Würzer et al., 2016).

The latter category, physically-based or process-based models, includes SNTHERM (Jordan, 1991), SNOWPACK (Lehning et al., 1999), iSnoB (Marks et al., 1999), SnowModel (Liston & Elder, 2006), and FSM (R. Essery, 2015), among others. Common applications of these models include avalanche forecasting, detailed snow physics studies, and mountain hydrology (as distinguished from the broader scale river basin hydrology applications of temperature index models) in plot scale to catchment sized domains. When process-based snow models are applied to continental or global domains, sometimes as part of land surface or climate models, high performance computing infrastructure is typically needed (e.g. (Garen & Marks, 2005; Ikeda et al., 2021; Walter et al., 2005; Wrzesien et al., 2018)). While process-based models require more forcing variables than temperature index models, they also provide more detailed snow simulations which may include snow stratigraphy, sublimation losses, surface albedo, etc. Since they are based on equations that directly represent snowpack physics, they are expected to be more robust under changing climatic conditions.

The SnowClim model (Lute et al., 2022) was developed to fill the gap outlined above. Namely, it provides high simulation accuracy, incorporates all major energy balance calculations, and is computationally efficient enough to be run at sub-kilometer scale over regional to continental domains. However, the original MATLAB-based model has no separation of I/O and, relies on .mat files, and MATLAB usage is tied to licensing issues, which restricts wider applicability. The Python translation of SnowClim, pySnowClim, makes the model more robust and accessible to a wider scientific and practical audience when compared to the original MATLAB-based model

Software design

In the original MATLAB SnowClim workflow, execution was driven by a wrapper script that prepared data (.mat files) and called the main routine. Parameters were loaded from a file, and the timestep loop computed physics while managing state and outputs. Thus, effectively orchestration and configuration were coupled with the core calculations. In addition, the state of the snowpack used a procedural approach which is more prone to state-update bugs. Additionally, while style works well for a MATLAB-only workflow, it makes it harder to embed SnowClim as a “callable component” inside another model’s time loop without the separation of I/O/configuration from computation.

pySnowClim adopts a layered architecture: a core physics engine (timestep calculations) is separated from a runner (I/O + orchestration) and a CLI for batch/operational use. The new implementation also prioritizes vectorized multi-point computation and standardized NetCDF/NumPy I/O, with explicit JSON-based configuration. Research workflows often need the same physics to run in different contexts: notebooks, batch jobs, or pipelines with standardized I/O. Separating the engine from I/O makes easier to validate physics independently of data inputs format.

pySnowClim uses an encapsulated state with more abstraction than the procedural approach

88 used in MATLAB, where the model state and outputs are explicit data structures (e.g.,
89 Snowpack state and preallocated SnowModelVariables).

90 Overall, these changes support maintainability, reduce risk of subtle state-update bugs, and
91 make easier the addition of new diagnostics, and variables. The changes allow pySnowClim to
92 be more easily coupled to hydrology/land-surface/ecosystem models as a drop-in component
93 (shared arrays/NetCDF, pipeline execution) without changing core physics.

94 Research impact statement

95 When a model is already published/validated (J. T. Anderson et al., 2025; Boeykens et al., 2025;
96 Dixon et al., 2026; Jans et al., 2025; Williams et al., 2026), researchers need confidence that
97 the new implementation is behaviorally comparable. Also, a well-supported Python package
98 enables easier integration, reproducibility, and further development. pySnowClim provides an
99 [extensive documentation](#) including API references, example datasets, and validation against
100 observations from a snow monitoring site. It also includes a comprehensive testing framework,
101 featuring unit tests for individual physics functions and integration tests for complete workflows.

102 In addition, the model is now usable as importable, tested, open-source Python software that
103 can be coupled into land-surface and hydrological modelling frameworks. For instance, there is
104 a current implementation of coupling between pySnowClim and the Community Water Model
105 (CWatM) (Burek et al., 2020) (<https://github.com/iiasa/CWatM/tree/develop>). The CWatM
106 documentation includes a dedicated [pySnowClim section](#), and notes that users can choose
107 between pySnowClim and the existing degree-day snow approach.

108 The model

109 pySnowClim employs the fundamental principles of mass and energy conservation as its core
110 framework. The model requires meteorological forcing data including temperature, precipitation,
111 shortwave radiation, longwave radiation, wind speed, humidity, and pressure to simulate critical
112 snow variables such as snow water equivalent (SWE), snow depth, snow density, snow melt,
113 snowpack liquid water content, albedo, and energy fluxes. [Figure 1](#) shows the snow model
114 conceptual diagram.

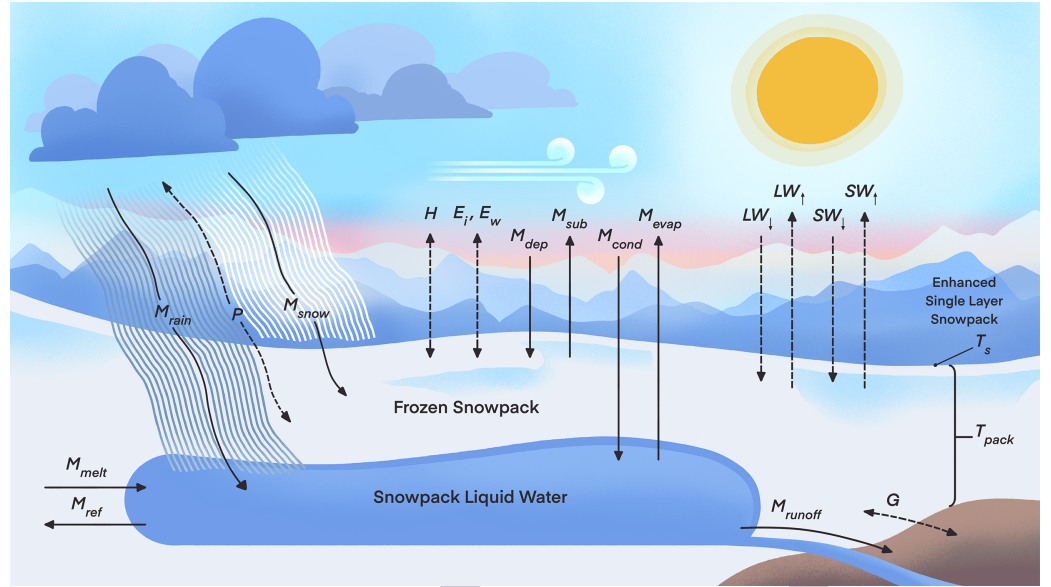


Figure 1: Snow model conceptual diagram. Solid black arrows indicate mass fluxes and dashed black arrows indicate energy fluxes. T_s is snow surface temperature and T_{pack} is the temperature of the snowpack.

pySnowClim calculates the net energy flux to the snow surface accounting for shortwave and longwave radiation, sensible and latent heat fluxes with stability corrections, ground heat flux, and precipitation heat flux Equation 1. The model is built around the surface energy balance equation:

$$Q_{net} = SW_{down} - SW_{up} + LW_{down} - LW_{up} + H + E_i + E_w + G + P \quad (1)$$

Where SW denotes shortwave radiation fluxes, LW denotes longwave radiation fluxes, H is the sensible heat flux, E is the latent heat flux of ice (i) and water (w), G is the ground heat flux, and P is the advected heat flux from precipitation.

The model tracks snow accumulation, sublimation and evaporation processes, snowmelt generation, and liquid water movement through the snowpack with mass conservation Equation 2. Mass balance of the solid (M_s) and liquid (M_l) portions of the snowpack are governed by:

$$M_s = M_{snow} + M_{ref} - M_{melt} + M_{dep} - M_{sub} \quad (2)$$

$$M_l = M_{rain} - M_{ref} + M_{melt} - M_{runoff} + M_{cond} - M_{evap} \quad (3)$$

Where M_{snow} is the mass of new snowfall, M_{ref} is the mass of the snowpack liquid water that has been refrozen, M_{melt} is the mass of snow that has melted, M_{dep} is the mass of deposition, M_{sub} is the mass of sublimation, M_{rain} is the mass of rain added to the snowpack, M_{runoff} is the mass of liquid water that has left the snowpack as runoff, M_{cond} is the mass of condensation, and M_{evap} is the mass of evaporation.

Other key model components include precipitation phase determination using bivariate logistic regression (Jennings et al., 2018), fresh snow density calculation based on temperature (E. A. Anderson, 1976), snow density evolution through compaction processes, liquid water retention and drainage. pySnowClim has three different albedo scheme implementations including aging

effects, melting conditions, and seasonal variations with options for different complexity levels (Richard Essery et al., 2013; Hamman et al., 2018; Liang et al., 1994; Tarboton & Luce, 1996). In addition, similarly to other models (E. Anderson, 2006; Burek et al., 2020), a new functionality was added to reduce excessive snow accumulation (i.e. snow towers) using an optional radiation enhancement.

AI usage disclosure

In preparing this work, the authors used ChatGPT (OpenAI) and Claude (Anthropic) to assist with the initial MATLAB-to-Python translation and generation of code documentation. The unit tests of the package were carried out by using Claude. After using these tools, the authors thoroughly reviewed, edited and refined the content, and take full responsibility for the content of this publication. No generative AI tools were used in the development of the writing of this manuscript.

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